

Vibe Audio Protocol (V.A.P.) v3.1

Universal Data Architecture & Taxonomy Definition

Executive Summary

The Vibe Audio Protocol (V.A.P.) is a 9-dimensional metadata standard designed to capture the holographic identity of an audio file. Unlike ID3 tags which capture *bibliographic* data (Who/What), V.A.P. captures *experiential* data (How/Why).

PILLAR 1: STRUCTURAL (The Skeleton)

Governs the mathematical, temporal, and dynamic architecture.

1.1 Temporal Dynamics

- **BPM (Raw):** The technical pulses per minute.
- **BPM (Perceived):** Half-time (0.5x), Double-time (2.0x), or Triplet-feel index.
- **Groove Quantization:**
 - *Machine Lock:* 100% rigid grid alignment (Techno/Trance).
 - *Human Swing:* Percentage of deviation (e.g., J Dilla Swing ~60%).
- **Time Signature:** Standard (4/4, 3/4) vs. Complex (7/8, 5/4).

1.2 Arrangement Architecture

- **Sectional Markers:** Timestamps for Intro, Verse, Chorus, Drop, Bridge, Outro.
- **Mix-Window Index:** Duration (bars) available for seamless DJ transitions.
- **Breakdown Depth:** Energy drop percentage during the bridge/breakdown.

1.3 Percussive DNA

- **Kick Transient Profile:**
 - *Attack:* Sharp (Click) vs. Soft (Thud).
 - *Decay:* Short (Tight) vs. Long (Boom/808).
- **Syncopation Index:** Ratio of on-beat to off-beat emphasis (0.0 - 1.0).
- **Ghost Note Density:** Frequency of low-velocity rhythmic subdivisions.

PILLAR 2: TONAL (The Flesh)

Governs the harmonic, melodic, and pitch-based content.

2.1 Harmonic Profile

- **Key Signature:** Root Note + Scale (Major, Minor, Dorian, Lydian, etc.).
- **Chord Complexity:**

- *Triadic*: Simple Root-3rd-5th (Pop/Rock).
 - *Extended*: 7ths, 9ths, 11ths, 13ths (Jazz/Neo-Soul).
- **Dissonance Rating**: Frequency of tension intervals (Minor 2nds, Tritones).

2.2 Melodic Contour

- **Range Span**: Octave spread between the lowest and highest melodic note.
- **Hook Strength**: Repetition frequency of the primary motif (Earworm Score).
- **Melodic Motion**: Conjunct (Step-wise) vs. Disjunct (Large leaps).

2.3 Tuning Standard

- **Reference Pitch**: 440Hz (Standard), 432Hz (Cosmic), 444Hz (Bright).
- **Microtonality**: Presence of non-western pitch bends or quarter tones.

PILLAR 3: TIMBRAL (The Skin)

Governs the texture, fidelity, and sonic physics.

3.1 Spectral Physics

- **Frequency Balance**:
 - *Sub-Dominant*: Energy < 60Hz.
 - *Mid-Forward*: Energy 500Hz - 2kHz (Vocal/Guitar focus).
 - *Air/Brilliance*: Energy > 10kHz.
- **Spectral Saturation**: Density of harmonics (Clean Sine vs. Distorted Square).

3.2 Production Aesthetic

- **Fidelity Score**:
 - *Lo-Fi*: Bit-crushing, limited bandwidth, noise floor.
 - *Hi-Fi*: Full spectrum, pristine transient response.
- **Dynamic Range (LRA)**: Loudness Range Average (Audiophile score).
- **Spatial Width**: Mono, Stereo, or Immersive (Binaural/Atmos).

3.3 Texture Grain

- **Surface**: Glassy, Gritty, Wooden, Metallic, Liquid.
- **Artifacts**: Vinyl crackle, Tape hiss, 60-cycle hum (intentional).

PILLAR 4: LINGUISTIC (The Voice)

Governs the semantic message and vocal delivery.

4.1 Semantic Content

- **Explicit Filter**: Clean, Mild, Explicit (Profanity), Severe (Hate/Violence).

- **Topic Clusters:** Love, Rebellion, Money, Depression, Party, Politics.
- **Narrative Arc:** Linear Story vs. Abstract/Repetitive Mantra.

4.2 Vocal Texture

- **Position:** Intimate (Close-mic), Distant (Reverb-soaked), buried.
- **Delivery Style:** Sung, Rapped, Screamed, Spoken, Whispered.
- **Processing:** Dry, Auto-Tuned, Chopped & Screwed, Vcoded.

4.3 Language Profile

- **Primary Language:** ISO 639-1 Code (en, es, fr, etc.).
- **Dialect/Slang:** Specific regional markers (e.g., Memphis Phonk, UK Drill).

PILLAR 5: AFFECTIVE (The Heart)

Governs the psychological and emotional impact.

5.1 The Thayer Coordinates

- **Valence:** Positivity (-1.0 to +1.0).
- **Arousal:** Energy Level (0.0 to 1.0).
- **Dominance:** Empowering/Aggressive vs. Submissive/Vulnerable.

5.2 Emotional Complexity

- **Mood Stability:** Constant state vs. Volatile shifts.
- **Catharsis Potential:** Suitability for emotional venting/release.
- **Nostalgia Trigger:** Probability of evoking past memories (Era-dependent).

5.3 Tension Arc

- **Build-up Velocity:** Rate at which tension increases.
- **Resolution State:** Triumphant, Melancholic, or Unresolved (Cliffhanger).

PILLAR 6: CONTEXTUAL (The Scene)

Governs environmental and situational tagging.

6.1 Scenario Engine

- **Macro-Setting:** Gym, Car, Bedroom, Club, Nature, Office.
- **Micro-Activity:** Night Drive, Heavy Lifting, Deep Work, Intimacy, Commute.
- **Social Setting:** Solo, Couple, Small Group, Crowd/Mass.

6.2 Intent Vectors

- **Functional Goal:** Focus, Hype, Sleep, Seduction, Filter (Weed-out).
- **Time of Day:** Morning, Golden Hour, Late Night, 3AM.

6.3 Meteorological Match

- **Weather:** Rain, Sun, Fog, Snow, Storm.
- **Temperature:** Cold (Digital/Isolation) vs. Warm (Analog/Soul).

PILLAR 7: PHOTOMETRIC (The Eye)

Governs visual data for lighting and syncing (Standardized).

7.1 Chromatic Map

- **Primary Hex:** Dominant color code (e.g., #191970).
- **Secondary Hex:** Accent color code.
- **Palette Temperature:** Cool (Blues/Purples) vs. Warm (Reds/Oranges).

7.2 Lumen Dynamics

- **Brightness Floor/Ceiling:** Minimum vs Maximum intensity (0.0 - 1.0).
- **Strobe Trigger:** Threshold for rapid light actuation.
- **Fade Rate:** Sharp (On/Off) vs. Smooth (Linear Fade).

7.3 Visual Texture

- **Fog Density:** Recommended haze level (0.0 - 1.0).
- **Laser Compatibility:** Suitability for beam effects.
- **Visual Noise:** Static/Glitch vs. Clean/Solid.

PILLAR 8: KINETIC (The Body)

Governs biometric and physical response data.

8.1 Biometric Entrainment

- **Target HR Zone:** BPM range for heart rate lock (e.g., 110-130).
- **HRV Impact:** Stress induction (Low HRV) vs. Flow induction (High HRV).
- **Breath Rate:** Recommended breathing cadence.

8.2 Motor Response

- **Drive:** Impulse to move forward (Run/Walk).
- **Sway:** Impulse to rock side-to-side.
- **Head-Nod:** Neck muscle activation index.

8.3 Energy Expenditure

- **MET Score:** Metabolic Equivalent of Task (Intensity of activity supported).

PILLAR 9: GENEALOGICAL (The Roots)

Governs historical lineage, sampling, and cultural tribe.

9.1 Era Anchoring

- **Release Date:** Official timestamp.
- **Cultural Era:** The perceived time (e.g., "Y2K," "Golden Era," "Baroque").
- **Timelessness Score:** How dated the production sounds (0.0 - 1.0).

9.2 DNA & Sampling

- **Sample Lineage:** Original sources used (e.g., "Amen Break").
- **Interpolation:** Melodies borrowed from history.
- **Genre Tree:** Parent Genre -> Sub-Genre -> Micro-Genre.

9.3 Tribe Alignment (GVS)

- **Subculture ID:** Juggalo, Goth, Raver, Audiophile, Metalhead.
- **Authenticity Score:** "True" vs. "Poser" rating.
- **Viral Velocity:** Trend status and meme potential.